Unit 5.5 Integration

Note Title 2016/05/18

Note:

- Anti-derivative (also indefinite integral) – $\int f(x)dx$ — function
- Definite integral – $\int_a^b f(x)dx$ — number
- Area – make a sketch – positive number.

Integral types:

I. $\int x^n . dx = \frac{1}{n+1} x^{n+1} + C.$

$\rightarrow \int (f(x))^n . f'(x) dx = \frac{1}{n+1} (f(x))^{n+1} + C$

Examples:

1. $\int \left(x^4 + \underbrace{\frac{1}{x^4}}_{x^{-4}}\right) dx = \frac{1}{5}x^5 + \frac{x^{-3}}{-3} + C.$

2. $\int 2x(x^2+1)^3 dx = \frac{1}{4}(x^2+1)^4 + C.$

3. $\int \left(\sqrt{x} + \overset{x^{-1/2}}{\frac{1}{\sqrt{x}}}\right) dx = \frac{2}{3}x^{3/2} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} + C$

$= \frac{2}{3}x^{3/2} + 2x^{1/2} + C.$

4 $\frac{1}{2}\int 2x(\underbrace{1+x^2}) ^4 dx = \frac{1}{10}(1+x^2)^5 + C.$

5. $\int x\sqrt{1+x^2}\,dx$ $= \underbrace{\frac{1}{2}\cdot\frac{2}{3}(1+x^2)^{3/2}}$

$= \frac{1}{3}(1+x^2)^{3/2} + C.$

II. $\int \sin ax\,dx = \frac{-1}{a}\cos ax + C.$ | $-\sin ax \times a$

$\int \underbrace{\cos x}\,.(\sin x)^n\,dx = \frac{1}{n+1}(\sin x)^{n+1} + C.$

$\int f'(x)\sin(f(x))\,dx = -\cos(f(x)) + C.$

Examples

1. $\int \sin\frac{x}{2}\,dx = -2\cos\frac{x}{2} + C$

2. $\int \sin x.\cos^3 x.dx = -\frac{1}{4}\cos^4 x + C$

3. $\int x^2.\sin(x^3+1).dx = \frac{-1}{3}\cos(x^3+1) + C$

4 $\int (\sin x)'\cos x.dx = \frac{1}{2}(\sin x)^2 + C$

5. $\int \sqrt{\sin 5x}\,.\cos 5x.dx = \frac{2}{15}(\sin 5x)^{3/2} + C$

$\frac{3}{2}(\sin 5x)^{1/2}\times 5$

I and II mixed examples

$$\int x^4 (1+x^5)^6 \, dx = \frac{1}{35}(1+x^5)^7 + C$$

$$\int (x+1)(x^2+2x)^5 \, dx = \frac{1}{12}(x^2+2x)^6 + C$$

$$\int \frac{1}{\sqrt{x}} \cdot \sin\sqrt{x} \cdot dx = -2\cos\sqrt{x} + C.$$

$$\int x \cdot \cos x^2 \cdot \sqrt{\sin x^2} \, dx.$$

$$\int (x - \sin 5x)^4 (1 - 5\cos 5x) \, dx$$

$$\int \cos^2 4x \cdot \sin 4x \cdot dx$$

III. $\int e^{ax}\,dx = \frac{1}{a}e^{ax} + c$

$$\int f'(x)\cdot e^{f(x)}\,dx = e^{f(x)} + c.$$

Examples

$$\int e^{3t}\,dt = \frac{1}{3}e^{3t} + c$$

$$\int x^2\cdot e^{x^3+1}\,dx = \frac{1}{3}e^{x^3+1} + c$$

$$\int e^{\tan 3x}\cdot\sec^2 3x\,dx = \frac{1}{3}e^{\tan 3x} + c$$

$$\int \sec^2(e^x)\cdot e^x\,dx = \tan(e^x) + c$$

$$\int \cos x\cdot e^{3\sin x}\,dx = \frac{1}{3}e^{3\sin x} + C.$$

IV. $$\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C.$$

Examples

$$\int \frac{1}{2x+1}\,dx = \frac{1}{2}\ln|2x+1| + C.$$

$$\int \frac{\cos x}{1+2\sin x}\,dx = \frac{1}{2}\ln|1+2\sin x| + C$$

Type I.

$$\int \frac{\cos x}{(1+2\sin x)^3}\,dx = -\frac{1}{4}(1+2\sin x)^{-2} + C.$$

$$\int \frac{\sec x\tan x + \cos x}{\sec x + \sin x}\,dx = \ln|\sec x + \sin x| + C$$

$$\int \frac{dx}{\sqrt{x}.\ln(\sqrt{x}+1)\times(\sqrt{x}+1)}$$

$$= \int \frac{1}{\ln(\sqrt{x}+1)} \times \frac{1}{\sqrt{x}+1} \times \frac{1}{\sqrt{x}}\cdot dx$$

$$= 2\ln|\ln(\sqrt{x}+1)| + C$$

$$\frac{d}{dx}\, 2\ln|\ln(\sqrt{x}+1)| = \cancel{2}\, \frac{1}{\ln(\sqrt{x}+1)} \times \frac{1}{\sqrt{x}+1} \times \frac{1}{\sqrt{x}}$$

V. $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + c$

or $\sin^{-1} x + c$.

$$\int \frac{f'(x)}{\sqrt{1-(f(x))^2}}\,dx = \arcsin f(x) + c.$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + c.$$

$$\int \frac{f'(x)}{1+(f(x))^2}\,dx = \arctan f(x) + c.$$

Examples

NB

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C$$

$$\int \frac{x}{\sqrt{1-x^2}}\,dx = \int x(1-x^2)^{-1/2}\,dx$$

$$= -(1-x^2)^{1/2} + c.$$

$$\int \frac{1}{1+x^2}\,dx = \arctan x + C$$

$$\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\ln|1+x^2| + C$$

$$\int \frac{x}{(1+x^2)^2}\,dx = -\frac{1}{2}(1+x^2)^{-1} + c.$$

$$\int \frac{x^2}{1+x^6}\,dx = \int \frac{x^2}{1+(x^3)^2}\,dx = \frac{1}{3}\arctan(x^3) + C.$$

$$\int \frac{e^{2x}}{\sqrt{1-e^{4x}}}\,dx = \int \frac{e^{2x}}{\sqrt{1-(e^{2x})^2}}\,dx = \frac{1}{2}\arcsin(e^{2x}) + C$$

$$\int \frac{\sec^2 x}{\sqrt{1-\tan^2 x}}\,dx = \arcsin(\tan x)$$

Two handy forms

$$\int \frac{1}{\sqrt{a^2-x^2}}\,dx = \arcsin \frac{x}{a} + C$$

$$\int \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\arctan \frac{x}{a} + C.$$

Examples

$$\int \frac{1}{\sqrt{9-x^2}}\,dx = \arcsin \frac{x}{3} + C$$

$$\int \frac{1}{4+x^2}\,dx = \frac{1}{2}\arctan \frac{x}{2} + C.$$

Three " plans ".

1. $\int \frac{x^4+1}{x^3}\,dx = \int \left(x + \frac{1}{x^3}\right) dx$ ✓

$\cancel{\frac{x^3}{x^4+1} = \frac{x^3}{x^4}} + \frac{x^3}{1}$ X

2. $\int \frac{x^2+1-1}{x^2+1}\,dx = \int \left(1 - \frac{1}{x^2+1}\right) dx$

3. $\int x^2(1+x^2)^2\,dx = \int x^2(1+2x^2+x^4)\,dx$
$= \int (x^2+2x^4+x^6)\,dx$

VI $\int a^x . dx = \frac{1}{\ln a} . a^x + C.$

$\int f'(x)\, a^{f(x)} . dx = \frac{1}{\ln a}\, a^{f(x)} + c.$

Examples

$$\int \cos x \cdot 2^{\sin x} \cdot dx = \frac{1}{\ln 2} 2^{\sin x} + C.$$

$$\int \operatorname{cosec} x \cdot \cot x \cdot 8^{\operatorname{cosec} x} \cdot dx = -\frac{1}{\ln 8} 8^{\operatorname{cosec} x} + C$$

$$\int \frac{\cos\sqrt{x} \cdot \boxed{3}^{(\sin\sqrt{x}+2)}}{\sqrt{x}} \cdot dx = \frac{2}{\ln 3} 3^{\sin\sqrt{x}+2} + C$$

$$\int \sec^2 (3^x + x^2) \cdot \underbrace{(3^x \cdot \ln 3 + 2x)}\cdot dx.$$

$$= \tan(3^x + x^2) + C.$$

VII $\int \sinh x \cdot dx = \cosh x + C$

$\int \cosh x \cdot dx = \sinh x + C.$

$\int \sinh(f(x)) \cdot f'(x)\,dx = \cosh(f(x)) + C.$

$\int \cosh(f(x)) \cdot f'(x)\,dx = \sinh(f(x)) + C.$

Examples

$\int e^{\sinh x} \cdot \cosh x \cdot dx = e^{\sinh x} + C$

$\int \frac{\sinh(\ln x)}{x} \cdot dx = \cosh(\ln x) + C$

$\int \cosh(3^x) \cdot 3^x \cdot dx = \frac{1}{\ln 3} \sinh(3^x) + C.$

Finally.

$$\int \frac{3^x}{\sqrt{1-3^x}}\,dx = \frac{-2}{\ln 3}\left(1-3^x\right)^{1/2} + C$$

$$\int \frac{3^x}{\sqrt{1-3^{2x}}}\,dx = \frac{1}{\ln 3}\arcsin\left(3^x\right) + C.$$